

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1799

Belleair 138 kV
December 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1799. The Interconnection Customer proposes a 202.5 MW battery energy storage facility interconnecting at the Belleair 138 kV substation in the Virginia area. The request was received on June 27, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 533 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,703,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
L GRAND – WEST WINCHESTER 138 KV	INTERNATIONAL PAPER VALLIANT OK – W WINCHESTER SUBSTATION 138 KV	137.5	257	8.7
ABBOTSFORD – LE GRAND 138 KV line	INTERNATIONAL PAPER VALLIANT OK – W WINCHESTER 138 KV	130.4	288	8.7
TERROR LAKE MICROGRID – KITIMAT 138 KV	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345 KV L	129.0	283	15.1
KITIMAT – CASCADE STEEL 138 KV L	KEARNEYVILLE – INTERNATIONAL PAPER VALLIANT OK 345 KV line	128.0	296	15.1
CASCADE STEEL – FEAGAN'S MILL 138 KV	KEARNEYVILLE – INTERNATIONAL PAPER VALLIANT OK 345 KV	126.6	310	15.1
COW – FEAGAN'S MILL 138 KV	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345 KV	126.6	315	15.1
YAMHILL – COW 138 KV line	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345 KV	125.8	328	15.1
YAMHILL – 3 MILE KNOLL 138 KV	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345 KV	125.3	335	15.1
E ELLENSBURG – WEST HILLS 138 KV	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345 KV Line	116.8	282	15.1
TERROR LAKE MICROGRID – E ELLENSBURG 138 KV	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE SUBSTATION 345 KV	116.0	296	15.1
SANTA ROSA ENERGY CENTER – MONOLITH 138 KV	INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345 KV	106.5	351	9.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BELLEAIR – WPCP 138 KV line	KEARNEYVILLE – INTERNATIONAL PAPER VALLIANT OK 345 KV	104.2	371	56.3
SHEPHERDSTOWN SUBSTATION – LCEC NO. 5 138 KV	CLARK COUNTY – KEARNEYVILLE 345 KV line	103.3	338	8.7
BELLEAIR SUBSTATION – OLD CHAPEL 138 KV	SANTA ROSA ENERGY CTR – BALCH 138 KV Line	103.0	395	53.5
OLD CHAPEL – HP HOOD CT 138 KV	SANTA ROSA ENERGY CTR – BALCH 138 KV	102.4	409	53.5
SPRINGFIELD – HP HOOD CT 138 KV	SANTA ROSA ENERGY CTR – BALCH 138 KV	101.7	429	53.5
COLUMBUS CITY 345/1kV Auto	CLARK COUNTY – KEARNEYVILLE 345 KV	100.5	563	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WESTWINCHESTER – LE GRAND 138 KV	Rebuild line with higher-capacity conductor	\$3,894,000	100% (\$3,894,000)
ABBOTSFORD – LE GRAND 138 KV	Rebuild line with higher-capacity conductor	\$8,407,000	100% (\$8,407,000)
TERROR LAKE MICRGRID – KITIMAT 138 KV	Rebuild line with higher-capacity conductor	\$4,308,000	44% (\$1,881,000)
CASCADE STEEL – KITIMAT 138 KV	Rebuild line with higher-capacity conductor	\$1,306,000	44% (\$570,000)
CASCADE STEEL – FEAGAN'S MILL 138 KV	Rebuild line with higher-capacity conductor	\$2,765,000	44% (\$1,207,000)
COW – FEAGAN'S MILL 138 KV	Rebuild line with higher-capacity conductor	\$2,902,000	44% (\$1,267,000)
YAMHILL – COW 138 KV	Rebuild line with higher-capacity conductor	\$2,826,000	44% (\$1,234,000)
YAMHILL – 3 MILE KNOLL 138 KV	Rebuild line with higher-capacity conductor	\$4,520,000	44% (\$1,974,000)
E ELLENSBURG – WEST HILLS 138 KV	Rebuild line with higher-capacity conductor	\$2,952,000	44% (\$1,289,000)
TERROR LAKE MICROGRID – E ELLENSBURG 138 KV	Rebuild line with higher-capacity conductor	\$3,185,000	44% (\$1,391,000)
SANTA ROSA ENERGY CENTER – MOOLITH 138 KV Line	Reconductor line	\$1,582,000	26% (\$408,000)
BELLEAIR SUBSTATION – WPCP 138 KV Line	Reconductor line	\$1,125,000	80% (\$902,000)
SHEPHERDSTOWN SUBSTATION – LCEC NO. 5 SUBSTATION 138 KV L	Reconductor line	\$668,000	15% (\$101,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
OLD CHAPEL – BELLEAIR 138 KV	Reconductor line	\$493,000	96% (\$473,000)
OLD CHAPEL – HP HOOD CT 138 KV	Reconductor line	\$3,741,000	96% (\$3,590,000)
HP HOOD CT – SPRINGFIELD 138 KV	Reconductor line	\$2,937,000	96% (\$2,818,000)
COLUMBUS CITY 345/1kV Transformer Bank	Replace transformer with higher-rated unit	\$27,510,000	12% (\$3,297,000)
	Total allocated to project		\$34,703,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.